ضمان جودة نظم المعلومات
12:00-1:00
30/11/2024
د/مروة حسين محمد

Faculty of Computers & Information, Assiut University
Course code: IS441
4th Level
Mid-term Exam
Duration: one hour
This exam for the following program(s) : (Add the program's name(s))

Points: -/15

1. الإسم الرباعي (بالعربي فقط) *

عمرو ايمن حسين احمد

2. رقم الجلوس *

162021225

3. المستوي *

○ الاول

- [ ] الثاني
- [ ] الثالث
- [ ] الرابع 2014
- [ ] الرابع 2015
- [ ] الرابع 2016
- [ ] الرابع 2017
- [ ] الرابع 2018
- [ ] الرابع 2019
- [ ] الرابع 2020
- [x] الرابع 2021

4. البرنامج *

- [x] عام

5. رقم المعمل *

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6. رقم الكمبيوتر *

16

7. الكود (قد تمت مراجعة بيانات الطالب ورقم الجلوس) *

NVQY

S **Will be reviewed**

8. Why is a DMP a "living document"?
K>

- It includes incomplete information
- It is discarded after the project ends
- It prioritizes only archived data
- (6) It evolves throughout the project lifecycle

S **Will be reviewed**

9. Which of the following is true about T-tests compared to ANOVA?

- ANOVA is used for paired data comparisons
- T-tests are used for two groups, while ANOVA is for three or more groups
- T-tests can handle more than two groups
- T-tests are always preferred over ANOVA

E3 **Will be reviewed**

10. Why are periodic backup verifications necessary?

- To identify duplicate data entries
- To ensure backups are encrypted
- To confirm data remains accessible and uncorrupted overtime

- To eliminate old files

S **Will be reviewed**

11. Which is an example of poor data entry?

- Consistent column names
- Consistent date formats
- Text and numbers in the same column
- Descriptive file names

S **Will be reviewed**

12. What is the main use case for a T-test? [5

- To test if sample data follows a normal distribution
- To check for correlations between two variables
- @) To compare means of small samples from two groups
- To compare the standard deviation of two datasets

O **Will be reviewed**

13. What is Kaggle primarily used for in scientific data management? CR

- Creating metadata
- Data cleaning
- Hosting datasets for analysis and machine learning

- ◯ Long-term data analysis

S Will be reviewed

14. What does data transformation involve? ER

- ◯ Visualizing unstructured datasets
- ◉ Cleaning, validating and preparing data for analysis
- ◯ Converting data into physical samples
- ◯ Developing hardware for datasets

S Will be reviewed

15. Which practice should researchers follow to ensure data integrity in the early stages of a project? EG,

- ◉ Automating data validation to catch errors early
- ◯ Discarding data with missing values without validation
- ◯ Not backing up data to save storage space
- ◯ Allowing only manual data entry without checks

O Will be reviewed

16. Which method would ensure consistency when recording pollutant levels from various sensors? EG>

- ◯ Using sensor data without metadata for simplicity
- ◯ Relying on manual data entry from printed reports
- ◉ Standardizing units (e.g., pg/m3) and recording formats across sensors

Allowing sensors to use their default units and formats

S **Will be reviewed**

17. How can researchers ensure the long-term accessibility of data in preservation? DCJ

- By storing data in obsolete formats
- By conducting regular integrity checks and migrating files to new formats
- By excluding metadata from archived datasets
- By limiting the number of data versions

S **Will be reviewed**

18. What hypothesis would you test with an independent samples t-test?

- The sample means are equal to zero
- The variances of two samples are equal
- The means of two independent groups are the same
- (6) The means of two related groups are the same

EZj **Will be reviewed**

19. What must be addressed when sharing sensitive data? EU

- Legal permissions and privacy concerns
- Data volume restrictions
- File naming conventions

Metadata standards

Will be reviewed

20. A biodiversity team uses citizen science platforms like iNaturalist to collect species observations. Which challenge is most likely encountered? ER

- Difficulty in generating graphs
- Limited number of participation from the public
- Inconsistent data formats and missing metadata
- Limited use of relational databases

Will be reviewed

21. What is data classification? ER

- Visualizing experimental results
- Organizing data based on sensitivity and importance
- Cleaning datasets
- A process to define metadata standards

Will be reviewed

22. What is a Data Management Plan (DMP)? ER

- A document outlining how data will be managed during and after research
- A financial plan for research funding
- A statistical analysis report

- ○ A legal document for data ownership

Will be reviewed

23. Why is data collection important?

- ◉ It ensures accuracy by collecting data in a structured and organized manner
- ○ It reduces the volume of datasets
- ○ It helps in building relational databases
- ○ It eliminates the need for metadata

Will be reviewed

24. Which of these scenarios would be appropriate for a one-sample t-test?

- ○ Testing the difference in salary between men and women
- ○ Comparing weight loss among different diet groups
- ◉ Comparing the average blood pressure of a group to a known value
- ○ Comparing test scores between two classes

Will be reviewed

25. What is the first stage of the Scientific Data Lifecycle?

- ◉ Plan
- ○ Preserve
- ○ Collect

Analyze

Will be reviewed

26. What is the primary goal of the scientific data lifecycle? ER

- Data analysis and visualization
- Ensuring data quality, accessibility, and preservation
- Funding for research
- Designing experiments

Will be reviewed

27. Which of the following is NOT part of assuring data quality? ER

- Accessibility
- Metadata creation
- Relevance
- Consistency

Will be reviewed

28. What does data organization aim to achieve? ER

- Development of proprietary storage systems
- Systematic arrangement for easy retrieval and interpretation
- Simplifying manual data entry

Creation of unstructured datasets

S **Will be reviewed**

29. What is imputation in data collection? ER

- Randomly guessing values for missing entries
- Using algorithms to predict and fill in missing values
- Avoiding data cleaning
- Deleting missing data

S **Will be reviewed**

30. Which tool is commonly used to track changes to both data and code in scientific research? ER

- GitHub (version control)
- Excel
- SQL
- Checksum algorithms

£3 **Will be reviewed**

31. What is statistical inference? ER

- A technique to sampling data
- The art of generating conclusions about data distribution
- A method to visualize data distributions

The process of summarizing raw data

Will be reviewed

32. Which of the following is **NOT** a key element of data assurance? ER

- Validation
- File compression
- Data integrity
- Data accuracy

Will be reviewed

33. Which of the following is **NOT** an example of scientific data? ER

- Experimental results
- Economia Data
- Social media posts and comments
- Climate Data

Will be reviewed

34. How do you interpret a p-value greater than 0.05 in hypothesis testing? DR

- All answers are correct
- Accept the null hypothesis
- Reject the alternative hypothesis

- ◯ Fail to reject the null hypothesis

S Will be reviewed

35. High data volume affects .... . [^J

- ◯ Ethical issues
- ◯ Metadata creation
- ◉ System performance
- ◯ Project timelines

S Will be reviewed

36. What is an effective naming convention for files?

- ◉ Use descriptive names including project and date information
- ◯ Avoid including the creation date in file names
- ◯ Use generic names like "data1.xls"
- ◯ Use names with spaces and special characters

O Will be reviewed

37. What could be the consequence of failing to validate climate data accuracy during research?

- ◯ Improved accuracy in forecasting
- ◯ Increased transparency in predictions
- (6) Misleading climate models and incorrect decisions

◯ Simplified data management

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